ACTIVIDAD SQOOP

Álvaro Fernández Becerra

Realiza la siguiente actividad queremos trasladar con Sqoop los datos de una tabla de una BBDD de MySQL llamada usuarios a nuestro Hive,
Tendremos que levantar un cluster EMR según los pasos indicados en Sqoop, conectarnos por consola y realizar los siguientes pasos. Primero usaremos Sqoop y luego entraremos a Hive para realizar la consulta.
Pon los comandos y las capturas en las zonas rosas de tal forma que salga tu nombre en el aula virtual arriba a la derecha y avísame cuando lo tengas hecho para verlo.

- Dada la siguiente cadena de conexión, el usuario `admin` y la contraseña `hola.1234`.

  `jdbc:mariadb://database.cjieg4mycwih.us-east-1.rds.amazonaws.com/usuarios`

- Mostramos las tablas con **sqoop list-tables** y el driver `--driver org.mariadb.jdbc.Driver`.

```
sqoop list-tables
--connect jdbc:mariadb://database.cjieg4mycwih.us-east-1.rds.amazonaws.com/usuarios
--username admin
--password hola.1234
--driver org.mariadb.jdbc.Driver
```

```
[hadoop@ip-172-31-28-187 ~]$ sqoop list-tables   --connect jdbc:mariadb://database.cjieg4mycwih.us-east-1.rds.amazonaws.
com/usuarios   --username admin   --password hola.1234   --driver org.mariadb.jdbc.Driver
Warning: /usr/lib/sqoop/../accumulo does not exist! Accumulo imports will fail.
Please set $ACCUMULO_HOME to the root of your Accumulo installation.
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/lib/hadoop/lib/slf4j-reload4j-1.7.36.jar!/org/slf4j/impl/StaticLoggerBinder.class
]
SLF4J: Found binding in [jar:file:/usr/lib/hive/lib/log4j-slf4j-impl-2.17.1.jar!/org/slf4j/impl/StaticLoggerBinder.class
]
SLF4J: Found binding in [jar:file:/usr/lib/hbase/lib/client-facing-thirdparty/slf4j-reload4j-1.7.33.jar!/org/slf4j/impl/
StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Reload4jLoggerFactory]
2026-03-19 02:01:42,062 INFO sqoop.Sqoop: Running Sqoop version: 1.4.7
2026-03-19 02:01:42,122 WARN tool.BaseSqoopTool: Setting your password on the command-line is insecure. Consider using -
P instead.
2026-03-19 02:01:42,248 WARN sqoop.ConnFactory: Parameter --driver is set to an explicit driver however appropriate conn
ection manager is not being set (via --connection-manager). Sqoop is going to fall back to org.apache.sqoop.manager.Gene
ricJdbcManager. Please specify explicitly which connection manager should be used next time.
2026-03-19 02:01:42,264 INFO manager.SqlManager: Using default fetchSize of 1000
usuarios
[hadoop@ip-172-31-28-187 ~]$
```

- Importamos la tabla **usuarios** en hive con **sqoop import**

```
sqoop import
  --connect jdbc:mariadb://database.cjieg4mycwih.us-east-1.rds.amazonaws.com/usuarios
  --username admin
  --password hola.1234
  --table usuarios
  --hive-import
  --driver org.mariadb.jdbc.Driver
```

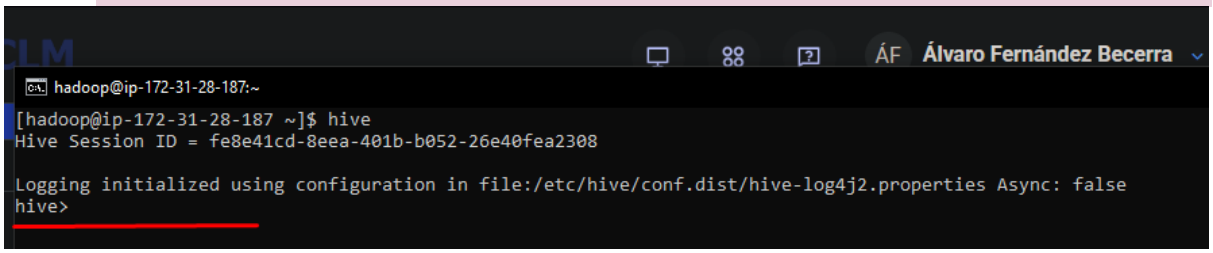
```
ve.ql.metadata.SessionHiveMetaStoreClient ugi=hadoop (auth:SIMPLE) retries=1 delay=1 lifetime=0
2026-03-19 02:05:48,030 INFO ql.Driver: Starting task [Stage-1:STATS] in serial mode
2026-03-19 02:05:48,031 INFO metastore.HiveMetaStoreClient: Closed a connection to metastore, current connections: 0
2026-03-19 02:05:48,032 INFO stats.BasicStatsTask: Executing stats task
2026-03-19 02:05:48,037 INFO metastore.HiveMetaStoreClient: Trying to connect to metastore with URI thrift://ip-172-31-2
8-187.ec2.internal:9083
2026-03-19 02:05:48,040 INFO metastore.HiveMetaStoreClient: Opened a connection to metastore, current connections: 1
2026-03-19 02:05:48,041 INFO metastore.HiveMetaStoreClient: Connected to metastore.
2026-03-19 02:05:48,041 INFO metastore.RetryingMetaStoreClient: RetryingMetaStoreClient proxy=class org.apache.hadoop.hi
ve.ql.metadata.SessionHiveMetaStoreClient ugi=hadoop (auth:SIMPLE) retries=1 delay=1 lifetime=0
2026-03-19 02:05:48,161 INFO stats.BasicStatsTask: Table default.usuarios stats: [numFiles=4, numRows=0, totalSize=1568,
 rawDataSize=0]
2026-03-19 02:05:48,162 INFO ql.Driver: Completed executing command(queryId=hadoop_20260319020547_cb17b20e-8488-4819-a7a
f-7a32864d2485); Time taken: 0.457 seconds
OK
2026-03-19 02:05:48,162 INFO ql.Driver: OK
2026-03-19 02:05:48,162 INFO ql.Driver: Concurrency mode is disabled, not creating a lock manager
Time taken: 0.642 seconds
2026-03-19 02:05:48,162 INFO CliDriver: Time taken: 0.642 seconds
2026-03-19 02:05:48,162 INFO conf.HiveConf: Using the default value passed in for log id: ac821785-cd01-46ff-a2cb-9a61d4
e559cc
2026-03-19 02:05:48,162 INFO session.SessionState: Resetting thread name to  main
2026-03-19 02:05:48,163 INFO conf.HiveConf: Using the default value passed in for log id: ac821785-cd01-46ff-a2cb-9a61d4
e559cc
WARN: The method class org.apache.commons.logging.impl.SLF4JLogFactory#release() was invoked.
WARN: Please see http://www.slf4j.org/codes.html#release for an explanation.
2026-03-19 02:05:48,177 INFO tez.TezSessionPoolManager: Closing tez session if not default: sessionId=ac821785-cd01-46ff
-a2cb-9a61d4e559cc, queueName=null, user=hadoop, doAs=true, isOpen=false, isDefault=false
2026-03-19 02:05:48,177 INFO client.TezClient: Shutting down Tez Session, sessionName=HIVE-ac821785-cd01-46ff-a2cb-9a61d
4e559cc, applicationId=application_1773884820692_0002
2026-03-19 02:05:48,181 INFO client.TezClient: Could not connect to AM, killing session via YARN, sessionName=HIVE-ac821
785-cd01-46ff-a2cb-9a61d4e559cc, applicationId=application_1773884820692_0002
2026-03-19 02:05:48,191 INFO tez.TezSessionState: Attemting to clean up resources for ac821785-cd01-46ff-a2cb-9a61d4e559
cc: hdfs://ip-172-31-28-187.ec2.internal:8020/tmp/hive/hadoop/_tez_session_dir/ac821785-cd01-46ff-a2cb-9a61d4e559cc-reso
urces; 0 additional files, 0 localized resources
2026-03-19 02:05:48,197 INFO session.SessionState: Deleted directory: /tmp/hive/hadoop/ac821785-cd01-46ff-a2cb-9a61d4e55
9cc on fs with scheme hdfs
2026-03-19 02:05:48,197 INFO session.SessionState: Deleted directory: /tmp/hadoop/ac821785-cd01-46ff-a2cb-9a61d4e559cc o
n fs with scheme file
2026-03-19 02:05:48,197 INFO hive.HiveImport: Hive import complete.
2026-03-19 02:05:48,200 INFO hive.HiveClientCommon: Export directory is contains the _SUCCESS file only, removing the di
rectory.
[hadoop@ip-172-31-28-187 ~]$
```

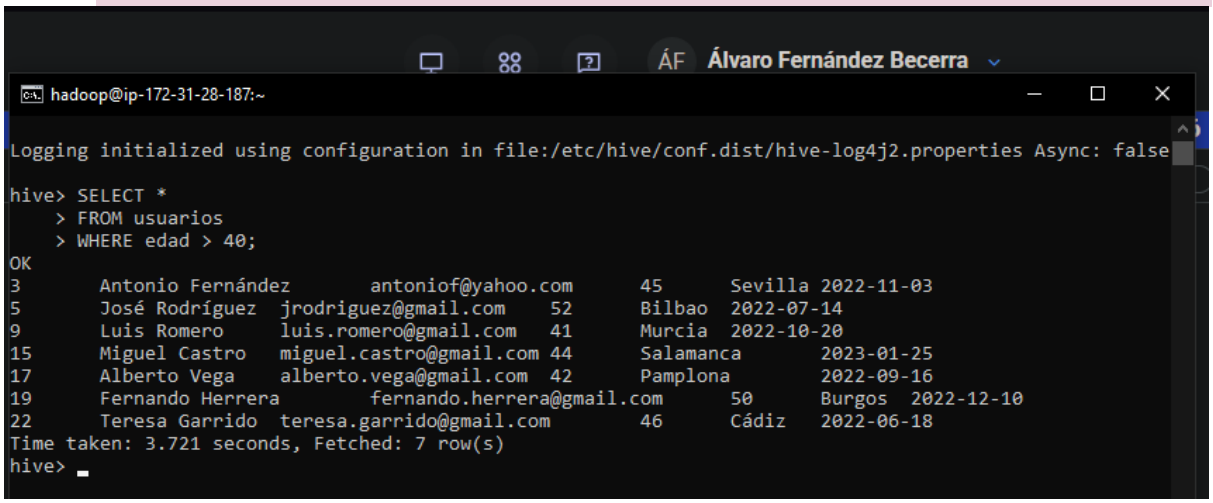
- Darle permisos al directorio /user de HDFS

  `hdfs dfs -chmod -R 777 /user`

- Nos conectamos a hive

```
hive
```



- Realizamos la siguiente consulta. Muestra los usuarios de más de 40 años.

```
SELECT *
FROM usuarios
WHERE edad > 40;
```